

A Comparative Study of Portfolio Insurance

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Abstract

This paper undertakes a comparative study of portfolio insurance under a variety of modeling strategies. Specifically, we focus on portfolio insurers who derive utility from horizon wealth, with marginal utility tending smoothly to infinity at some pre-specified floor. We solve for the optimal consumption-portfolio-wealth of these portfolio insurers and compare with “constrained” portfolio insurers and “normal agents.” General equilibrium conditions are contrasted under pure-exchange and production-type models. While the market price level is unambiguously increased under pure-exchange, under production the effect on market level is state-dependent. In both models the market volatility and risk premium are decreased by portfolio insurance. The paper also investigates the possible relationship between portfolio insurance type trading strategies and market volatility.

KEYWORDS: portfolio insurance, pure-exchange, production, volatility, trend-chasing.

JEL: C60, D51, D90, G11, G12.

1. Introduction

A portfolio insurance trading strategy is defined [Grossman and Vila (1989), Luskin (1988, p. 77)] as one which guarantees a minimum level of wealth (the “floor”) at some specified horizon, yet also participates in the potential gains of some reference portfolio (e.g., the S&P 500). Currently, the two most popular strategies implementing portfolio insurance are the synthetic put approach [Rubinstein and Leland (1981)] and the constant proportion portfolio insurance (CPPI) approach [Black and Jones (1987), Black and Perold (1992)].

Dynamic trading strategies following portfolio insurance are widely believed to affect the stock market level and volatility [Brady Report (1988)]. Indeed the primary focus of much recent work [Brennan and Schwartz (1989), Donaldson and Uhlig (1993), Basak (1995), Grossman and Zhou (1996)] has been to evaluate theoretically the effects on prices of the presence of portfolio insurance in the economy. Of this work, Grossman and Vila (1989) (partial equilibrium), Basak (1995), Grossman and Zhou (1996) (general equilibrium), have modeled portfolio insurers as expected utility maximizers, by explicitly applying to their expected utility maximization problem the additional constraint that the wealth at the portfolio insurance horizon be above the floor. The associated optimal portfolio policy endogenously justifies a strategy closely resembling the synthetic put approach to portfolio insurance.

One objective of this paper is to address some possible criticism of the existing modeling strategies. We provide a comparative analysis of alternative models of portfolio insurance: alternative approaches to embedding portfolio insurance objectives into optimal behavior; direct analysis of prescribed portfolio insurance trading strategies; and employing a production versus a pure-exchange environment as the basis for equilibrium. A by-product of this paper is that the portfolio insurance application highlights the critical differences and similarities between two workhorse models in dynamic asset pricing theory: the pure-exchange and production models.

A shortcoming of the previous expected utility maximizing formulation of portfolio insurance is that when the portfolio insurer’s horizon wealth hits the floor, his marginal indirect utility of wealth jumps discontinuously to infinity. It might seem more reasonable for an investor to become gradually, rather than suddenly, unhappy as his wealth approaches the floor. This leads us to the alternative formulation of portfolio insurance adopted in this paper, in which we let the portfolio insurers derive utility from wealth at the portfolio insurance horizon, and the utility function is chosen to implicitly endogenize the condition of wealth above the floor. Specifically, we let the marginal utility go smoothly to infinity at the floor so that the optimal horizon wealth automatically lies above the floor. The analysis reveals a strong similarity between the optimal trading strategy of these portfolio insurers and that of CPPI, providing justification for

the realism as well as reasonableness of this formulation. The strongest motivation, however, is the tractability gained by eliminating the cumbersomeness of an explicit constraint. The utility function we employ allows for a more elegant analysis; in particular we are able to extend beyond logarithmic preferences, to provide closed form solutions for agents exhibiting constant relative risk aversion (CRRA) preferences over intertemporal consumption.

We first look at the implications of portfolio insurance in a continuous-time version of the Lucas (1978) pure-exchange economy, populated by such portfolio insurers and “normal agents,” using martingale methods [Cox and Huang (1989), Harrison and Kreps (1979), Karatzas, Lehoczky, and Shreve (1987)]. Although the portfolio insurers here differ in the way they achieve their minimum level of horizon wealth, the Basak (1995) implications of a reduced market volatility and risk premium are reinforced (contrary to popular belief and related papers¹), with similar intuition. Briefly, portfolio insurers would like to go long in the (fixed supply) bond and hence the market volatility decreases to create extra incentive to invest in the risky market. Furthermore, due to their desire to achieve the floor at the portfolio insurance horizon, portfolio insurers want to forego pre-horizon consumption. However, aggregate consumption may not be postponed since its supply is exogenous (and perishable) in the pure-exchange model. To counteract the portfolio insurers’ demands, value of aggregate horizon wealth and aggregate consumption is increased, relative to an economy with no portfolio insurers. This leads to a higher pre-horizon market price in the presence of portfolio insurance.

This intuition motivates us to consider portfolio insurance in a production economy in which aggregate consumption is endogenous and so may actually be postponed. Moreover, the intuition for the reduced volatility motivates a model in which the supply of riskless bond is endogenous. A priori, the asset pricing implications could be quite different. We show (within a Cox-Ingersoll-Ross (1985) type production economy) that the reduced market volatility and risk premium are robust to this change of modeling, while the price level is now impacted very differently. It is now the aggregate consumption and not its valuation which is impacted by portfolio insurance: the pre-horizon aggregate consumption is unambiguously lower than in a normal-agent economy. We no longer conclude that the presence of portfolio insurance unambiguously increases the pre-horizon market price level. In “bad” states, in the sense that agents expect a relatively low horizon wealth, the market price is increased by the presence of portfolio insurance, but in “good”

¹Brennan and Schwartz (1989) and Grossman and Zhou (1996) argue that the market volatility is increased by the presence of portfolio insurers in a pure-exchange economy, in which agents are modeled as having objectives only over wealth at some portfolio insurance horizon. Our work is complementary to theirs in that we study portfolio insurance in a different economy, in which agents additionally derive utility over intertemporal consumption. A benefit of our having intertemporal consumption and hence the benchmark economy being Lucas (1978) is considerable tractability. In contrast to Brennan-Schwartz and Grossman-Zhou, who have to resort to numerical techniques in their analysis, our conclusions are deduced through explicit formulae for all quantities of interest.

states the market price is decreased. Since agents' valuation of consumption is exogenous, the portfolio insurers must shift consumption from good paths to bad paths to prevent their horizon wealth ever falling below the floor. Hence, the mechanism for the reduced market volatility is different from that in the pure-exchange model. The analysis provides a valuable illustration of the difference between a production and a pure-exchange model.

Finally, in light of the fact that a higher volatility due to portfolio insurance is commonly attributed to the “trend-chasing” behavior of such investment strategies, we analyze the trading behavior and its effect on volatility, of our optimizing portfolio insurers and also of some of the prescribed portfolio insurance trading strategies followed in practice. (A trend-chasing strategy is one in which funds are transferred to the risky assets as their prices go up and away from the risky assets as their price goes down.) Our overall conclusion is that within a traditional, Walrasian general equilibrium model there is no clear-cut relationship between the trend-chasing or contrarian nature of the portfolio insurers and the effect of their presence on market volatility. Microstructure liquidity considerations may be required to capture the connection.

Section 2 presents the multi-agent, continuous-time formulation and analyzes portfolio insurance at partial equilibrium. Section 3 investigates the general equilibrium implications in a pure-exchange economy, while Section 4 analyzes the production economy. Section 5 looks at the relationship between market volatility and the dynamic trading behavior of (optimizing and prescribed-strategy) portfolio insurers. Section 6 concludes. The Appendix provides all proofs.

2. General Formulation

This section describes the continuous-time framework for the pure-exchange (Sections 3 and 5) and the production (Section 4) economies to be analyzed subsequently. We consider a continuous-time, finite horizon $[0, T]$ economy populated by normal agents and portfolio insurers. There is a single consumption good serving as the numeraire. All the uncertainty in the economy is represented by a 1-dimensional Brownian motion, W , on a complete probability space $(\Omega, \mathcal{F}, \mathcal{P})$. Let $\{\mathcal{F}_t; t \in [0, T]\}$ be the augmentation by null sets of the filtration generated by W .

2.1. Investment Opportunities

We assume there are two investment opportunities: one instantaneously riskless and the other risky. In the equilibria of Sections 3 and 4, it will be shown that the vector of instantaneous net returns on the investment opportunities follows the dynamics

$$\begin{pmatrix} r(t) dt \\ \mu(t) dt + \sigma(t) dW(t) \end{pmatrix}, \quad (2.1)$$

where the interest rate r , the drift μ and the volatility σ are (possibly path-dependent) bounded, $\{\mathcal{F}_t\}$ -progressively measurable processes, with σ bounded away from zero.

Given the net return vector (2.1), market completeness allows us to construct a unique system of Arrow-Debreu securities. Accordingly, we define the *state price density process* ξ by

$$d\xi(t) = -\xi(t)[r(t) dt + \theta(t) dW(t)] , \quad (2.2)$$

where θ is the unique bounded, $\{\mathcal{F}_t\}$ -progressively measurable *market price of risk process*, given by $\theta(t) \equiv \sigma(t)^{-1}[\mu(t) - r(t)]$. $\xi(0)$ may be chosen arbitrarily. $\xi(t, \omega)$ is interpreted as the Arrow-Debreu price per unit probability $\mathcal{P}$ of one unit of consumption good in state $\omega \in \Omega$ at time t .

2.2. Agents' Endowments and Preferences

We assume there are N agents in the economy, the first M being portfolio insurers and the remaining $N - M$ normal agents. Each agent, n , is endowed at time zero with an initial wealth (in units of consumption good) of x_{n0} .

Each agent n chooses a nonnegative consumption process, c_n , and a 2-dimensional portfolio process, $\phi_n \equiv (\phi_{n0}, \phi_{n1})^T$, from the set of $\{\mathcal{F}_t\}$ -progressively measurable processes satisfying $E \left[\int_0^T c(t)^2 dt \right] < \infty$ and $E \left[\int_0^T \phi_{n1}(t)^2 X_n(t)^2 dt \right] < \infty$, and a nonnegative $\mathcal{F}_T$ -measurable horizon wealth $X_n(T)$ such that $E[X_n(T)^2] < \infty$. Here ϕ_n denotes the vector of fractions of agent n 's wealth, X_n , invested respectively, in the riskless and risky investment opportunities. Then, the n th agent's pre-horizon wealth process follows

$$dX_n(t) = X_n(t)r(t) - c_n(t) dt + X_n(t) \phi_{n1}(t)[\mu(t) - r(t)]dt + X_n(t) \phi_{n1}(t) \sigma(t) dW(t) . \quad (2.3)$$

It is well-known that the square-integrability condition on the portfolio process is sufficient to rule out arbitrage opportunities [Harrison and Pliska (1981)]. All the assumed integrability conditions are satisfied in the constructed equilibrium.

Given the dynamics of the state price density (2.2) and agents' wealth (2.3), an agent's wealth represents the cost of his future consumption stream plus horizon wealth, as stated in Lemma 2.1.

Lemma 2.1. *The wealth of agent n at time t is given by*

$$X_n(t) = \frac{1}{\xi(t)} E \left[\int_t^T \xi(s) c_n(s) ds \mid \mathcal{F}_t \right] + \frac{1}{\xi(t)} E[\xi(T) X_n(T) \mid \mathcal{F}_t] , \quad t \in [0, T] . \quad (2.4)$$

We model the agents as deriving utility $u_n(c_n(t))$ from consumption in $[0, T]$ and utility $v_n(X_n(T))$ from wealth at time T . We assume that all agents have power utility of consumption, $u_n(c) = c^\gamma / \gamma$, $\gamma < 1$, $\gamma \neq 0$ for $n = 1, \dots, N$. The portfolio insurers and normal agents differ,

however, in their utility of horizon wealth. Following the definition of portfolio insurance given in the Introduction, the portfolio insurers have a *modified* power utility function of time- T wealth, which implicitly inherits their desire to have a horizon wealth above the floor, with the marginal utility going smoothly to infinity at the floor, i.e.,

$$v_m(X) = \frac{(X - K)^\gamma}{\gamma}, \quad m = 1, \dots, M.$$

To make the normal agents comparable with the portfolio insurers, we assume that the normal agents have the standard power utility of time- T wealth, i.e.,

$$v_n(X) = \frac{X^\gamma}{\gamma}, \quad n = M + 1, \dots, N.$$

The portfolio insurers' preferences over horizon wealth exhibit decreasing relative risk aversion, and are more risk averse than those of the normal agents over the region $X(T) > K$.

Remark 2.1. In the interest of expositional simplicity, we do not explicitly model the future beyond time T , after the expiration of portfolio insurance. The future is summarized by the function $v(\cdot)$ defined over horizon wealth. As will be clear from the sequel, the exact assumptions made about the future beyond T do not influence our main results on market price, volatility and risk premium. Perhaps the most natural interpretation of $v(\cdot)$ is that of a commonly employed bequest function. At time T , agents in the economy hand their terminal wealth on to their heirs, resulting in the value of the heirs' initial endowment being $\xi(T)X(T)$. Alternatively, one can interpret this function as an indirect utility of post-portfolio insurance horizon wealth. Agents in the economy continue to live (and consume) beyond time- T , and their post-horizon consumption and utility thereof are summarized by the (indirect) utility function $v(\cdot)$. If the post-horizon period were to be modeled, the specific future utility functions of post-horizon consumption, as well as investment opportunities, would then be specified to be consistent with our assumed preferences of horizon wealth (details available upon request).²

2.3. Agents' Optimization Problems

We solve each agent's dynamic optimization problem using the martingale representation approach, in which the problem is converted into the following static variational one [Cox and Huang (1989), Karatzas, Lehoczky and Shreve (1987)]:

$$\max_{(c_n, X_n(T))} E \left[\int_0^T u_n(c_n(t)) dt + v_n(X_n(T)) \right] \quad \text{s.t.} \quad E \left[\int_0^T \xi(t) c_n(t) dt + \xi(T) X_n(T) \right] \leq \xi(0) x_{n0}.$$

²Yet another interpretation of the function $v(\cdot)$ is that of a direct utility defined over terminal lump-sum consumption as in Brennan and Schwartz (1989) and Grossman and Zhou (1996). With this interpretation, to attain equilibrium in our pure-exchange formulation (Section 3), the risky stock would additionally have to pay a liquidating dividend. All our conclusions regarding the market price and dynamics remain valid with the liquidating dividend extension.

Lemma 2.2 characterizes the optimal solution, assuming the feasibility condition $\xi(0)x_{m0} > KE[\xi(T)]$, $m = 1, \dots, M$.

Lemma 2.2. *Assume ξ is such that the expectations in (2.9) are well-defined. Then the optimal consumption processes and optimal time- T wealth of the N agents are*

$$\hat{c}_m(t) = [y_m \xi(t)]^{\frac{1}{\gamma-1}}, \quad m = 1, \dots, M, \quad (2.5)$$

$$\hat{c}_n(t) = [y_n \xi(t)]^{\frac{1}{\gamma-1}}, \quad n = M+1, \dots, N, \quad (2.6)$$

$$\hat{X}_m(T) = [y_m \xi(T)]^{\frac{1}{\gamma-1}} + K, \quad m = 1, \dots, M, \quad (2.7)$$

$$\hat{X}_n(T) = [y_n \xi(T)]^{\frac{1}{\gamma-1}}, \quad n = M+1, \dots, N, \quad (2.8)$$

where the y_m and y_n are the unique nonnegative numbers such that the budget constraints hold with equality, i.e.,

$$y_m = \left\{ \frac{E \left[\int_0^T \xi(t)^{\frac{\gamma}{\gamma-1}} dt + \xi(T)^{\frac{\gamma}{\gamma-1}} \right]}{\xi(0)x_{m0} - KE[\xi(T)]} \right\}^{1-\gamma}, \quad y_n = \left\{ \frac{E \left[\int_0^T \xi(t)^{\frac{\gamma}{\gamma-1}} dt + \xi(T)^{\frac{\gamma}{\gamma-1}} \right]}{\xi(0)x_{n0}} \right\}^{1-\gamma}. \quad (2.9)$$

Consequently: (i) if $x_{m0} = x_{n0}$, then $y_m > y_n$.

From (2.5)–(2.6) and property (i), relative to an identically endowed normal agent, a portfolio insurer foregoes consumption in $[0, T]$ in return for ensuring his time- T wealth lies above K . As required by the definition of portfolio insurance, the portfolio insurers also participate in the gains of the market, $1/\xi(T)$. (We later verify that the market is directly related to $1/\xi(T)$.) The optimal horizon wealth of a portfolio insurer equals the horizon wealth of a fictitious normal agent with “weight” y_m plus a riskless amount K . We can contrast this with the constrained portfolio insurers of Basak (1995) who purchase instead an additional put option. When the market is low, the horizon wealth of both types of portfolio insurers is higher than of an identically endowed normal agent, and vice versa when the market is high.

We report in Lemma 2.3 the portfolio strategies financing the optimal consumption processes and horizon wealth. Assuming ξ to follow a geometric Brownian motion in $[0, T]$ (consistent with the equilibria of Section 3 and 4), the portfolio insurers invest a smaller fraction in the risky asset than the normal agents, before the horizon.

Lemma 2.3. *Assume r and θ are constants. Agents’ optimal risky portfolio processes are*

$$\hat{\phi}_{m1}(t) = \frac{1}{(1-\gamma)} \left[1 - \frac{K \exp\{-r(T-t)\}}{\hat{X}_m(t)} \right] \left(\frac{\mu(t) - r}{\sigma(t)^2} \right), \quad m = 1, \dots, M, \quad (2.10)$$

$$\hat{\phi}_{n1}(t) = \frac{1}{(1-\gamma)} \left(\frac{\mu(t) - r}{\sigma(t)^2} \right), \quad n = M+1, \dots, N. \quad (2.11)$$

Hence, $\hat{\phi}_{m1}(t) \leq \hat{\phi}_{n1}(t)$.

For analytical convenience, we define a representative agent's utility of consumption by

$$U(c; \Lambda) \equiv \max_{c_1, \dots, c_N} \sum_{n=1}^N \lambda_n u_n(c_n) ; \quad \Lambda = (\lambda_1, \dots, \lambda_N) \in (0, \infty)^N ,$$

subject to $c_1, \dots, c_N \geq 0$ and $\sum_{n=1}^N c_n = c$. The representative agent's indirect utility of time- T wealth, $V(X; \Lambda)$, is defined analogously. Due to our choice of the individual agents' utility functions, the representative agent behaves as a portfolio insurer with floor MK :

$$U(c; \Lambda) = g(\Lambda) \frac{c^\gamma}{\gamma} , \quad V(X; \Lambda) = g(\Lambda) \frac{(X - MK)^\gamma}{\gamma} ,$$

where $g(\Lambda) \equiv (\sum_{n=1}^N \lambda_n^{1/(1-\gamma)})^{1-\gamma}$. His preferences depend only on the number of portfolio insurers in the economy and not on their individual initial endowments. This goes hand-in-hand with the well-known fact that our assumed preferences satisfy the aggregation property [Rubinstein (1974)], in the sense that equilibrium prices are determined independently of the distribution of initial endowments. This aggregation property of the agents' preferences of both consumption and time- T wealth makes the equilibrium analysis of a heterogeneous-agent economy more tractable in this alternative formulation of portfolio insurance than in Basak (1995), allowing in particular the extension from logarithmic to general CRRA preferences over consumption.

Remark 2.2 (Extensions).

- (i) The extension to multiple risky investment opportunities is straightforward and does not change the conclusions stated in the propositions and corollaries of Sections 3–4.
- (ii) If agents had log utilities of consumption and final wealth, $u_m(c) = u_n(c) = \log(c)$, $v_m(c) = \log(X - K)$ and $v_n(X) = \log(X)$, we would deduce all stated results with $\gamma = 0$.
- (iii) Portfolio insurers could have differing floors, K_m ; MK would simply be replaced by $\sum_{m=1}^M K_m$.

3. Equilibrium in a Pure-Exchange Economy with Portfolio Insurance

This section adapts a continuous-time variation on the Lucas (1978) pure-exchange economy. For recent continuous-time, general equilibrium models of this type, see Duffie and Huang (1985), Duffie (1986), Huang (1987), Duffie and Zame (1989), and Karatzas, Lehoczky, and Shreve (1990, 1991). In this economy, the investment opportunities faced by the agents are financial securities. The riskless investment is a bond in zero net supply; the risky investment is a risky stock in constant net supply of 1 and paying out a dividend stream at rate δ . We assume the (exogenously given) dividend process to follow a geometric Brownian motion process:

$$d\delta(t) = \delta(t) [\mu_\delta dt + \sigma_\delta dW(t)] ,$$

with $\mu_\delta, \sigma_\delta$ constants and $\delta(0) > 0$. Then the vector (2.1) describes the dynamics of the return (capital gains plus dividend payment) on the two securities, i.e.,

$$\begin{pmatrix} dP_0(t)/P_0(t) \\ (dP_1(t) + \delta(t) dt)/P_1(t) \end{pmatrix} = \begin{pmatrix} r(t) dt \\ \mu(t) dt + \sigma(t) dW(t) \end{pmatrix}.$$

In this case, the parameters r, μ and σ in (2.1) are endogenous and are determined in equilibrium.³ We endow each agent with e_n units (exogenous) of the risky asset, so that the agent's endogenous initial wealth is $x_{n0} = P_1(0) e_n$.

Definition 3.1 of equilibrium enforces clearing in the good and both security markets.

Definition 3.1. *An equilibrium in a pure-exchange economy is a collection of (r, μ, σ) and optimal $(\hat{c}_n, \hat{\phi}_n, \hat{X}_n(T))$, such that*

$$\sum_{n=1}^N \hat{c}_n(t) = \delta(t), \quad \sum_{n=1}^N \hat{\phi}_{n1}(t) \hat{X}_n(t) = P_1(t), \quad \sum_{n=1}^N \hat{X}_n(t) = P_1(t). \quad (3.1)$$

3.1. Equilibrium State Prices and Dynamics

In Proposition 3.1 we make use of the representative agent constructed in Section 2.3, where the equilibrium $g(\Lambda)$ is set equal to 1, without loss of generality.

Proposition 3.1. *Equilibrium exists and the equilibrium state price density is given by*

$$\xi(t) = U'(\delta(t); \Lambda) = \delta(t)^{\gamma-1}, \quad t \in [0, T], \quad (3.2)$$

where $\Lambda = (1/y_1, \dots, 1/y_N)$, and the y 's satisfy (2.9) with (3.2) substituted in. The equilibrium interest rate and market price of risk are constants given by

$$r = (1 - \gamma)\mu_\delta - \frac{(1 - \gamma)(2 - \gamma)}{2} \sigma_\delta^2 \quad \text{and} \quad \theta = (1 - \gamma)\sigma_\delta.$$

We conclude that ξ, r and θ are not affected by portfolio insurance. This is because it is the agents' weights alone that capture the effects of portfolio insurance, and these weights appear only as a multiplicative factor in the representative agent's marginal utility, and hence in ξ .

³ $P_1(T)$ is not restricted to be zero since the risky asset is allowed to be long-lived, still paying dividends after the portfolio insurance has expired. The pre-horizon stock value $P_1(t)$, $t \leq T$, coincides with the market portfolio value, which is determined in Section 3.2. On the other hand, the post-portfolio insurance horizon value of the stock would be given by the equilibrium valuation of the post-horizon dividend stream (if it were specified).

3.2. Market Price, Volatility and Risk Premium

The price of the *market* portfolio, $X_{MKT}(t)$, is defined as the aggregate optimally invested wealth in the investment opportunities, i.e.,

$$X_{MKT}(t) \equiv \sum_{n=1}^N \hat{\phi}_{n1}(t) \hat{X}_n(t) = \sum_{n=1}^N \hat{X}_n(t) = P_1(t) .$$

The equilibrium market price dynamics can be represented by

$$dX_{MKT}(t) + \delta(t) dt = X_{MKT}(t) [\mu_{MKT}(t) dt + \sigma_{MKT}(t) dW(t)] ,$$

so that μ_{MKT} is the *market drift*, $|\sigma_{MKT}|$ the *market volatility*, and $\mu_{MKT} - r$ the *market risk premium*. Propositions 3.2 – 3.3 provide analytical expressions for the equilibrium market price, risk premium and volatility, in terms of quantities all expressed in terms of exogenous parameters.

Proposition 3.2. *In equilibrium, the market price is*

$$X_{MKT}(t) = a(t) \delta(t) + b(t) \delta(t) + MK \exp\{-r(T-t)\} , \quad (3.3)$$

where $a(t) \equiv [\exp\{\eta(T-t)\} - 1]/\eta \geq 0$, $b(t) \equiv \exp\{\eta(T-t)\}$, $\eta \equiv \gamma \mu_\delta - \gamma(1-\gamma)\sigma_\delta^2/2$, and r is as given in Proposition 3.1. Consequently: (i) $X_{MKT}(t)$ is increasing in M or K ; (ii) $X_{MKT}(t)$ is increasing in $\delta(t)$ (within an economy, i.e., for a fixed M and K).

The market price in the portfolio insurance economy equals that in a normal agent ($M = 0$) economy plus M positive riskless terms $K \exp\{-r(T-t)\}$. Hence the market price is unambiguously increased by the presence of portfolio insurance, guaranteeing at the horizon the minimum level demanded by portfolio insurers, MK . The first component in (3.3), $a(t) \delta(t)$, is the current value of pre-horizon dividends, and is independent of portfolio insurance. The second plus third components, $b(t) \delta(t) + MK \exp\{-r(T-t)\}$, are the current value of the horizon market portfolio (the current value of post-horizon dividends), and increase in M and K . Due to their desire to achieve the floor at the portfolio insurance horizon, portfolio insurers want to forego pre-horizon consumption. However, aggregate consumption may not be postponed since its supply is exogenous and perishable. Hence, to counteract the portfolio insurers' demands, value of aggregate horizon wealth and aggregate consumption increases. Explained in terms of our specific utility of horizon wealth, the portfolio insurers always have a higher marginal utility for horizon wealth than the normal agents and so derive extra (indirect) marginal utility from post-horizon consumption. The tractability of the utility-of-horizon-wealth approach allows crisp comparative statics. The increase in market value becomes more pronounced as the number of portfolio insurers increases or as their floor increases, but does not depend on their initial endowments. The

higher the floor, the harder it is for a portfolio insurer to meet his portfolio insurance condition, and so the more he values future post-horizon consumption. Result (ii) reveals that $X_{MKT}(t)$ is directly related to $\delta(t)$, which in turn (by (3.2)) is directly related to $1/\xi(t)$. In particular, the market at time T is directly related to $1/\xi(T)$, as we anticipated in Section 2.

Proposition 3.3 presents the market price return dynamics.

Proposition 3.3. *The equilibrium market price return volatility and risk premium are*

$$|\sigma_{MKT}(t)| = \left(1 - \frac{MK \exp\{-r(T-t)\}}{X_{MKT}(t)}\right) |\sigma_\delta|, \quad (3.4)$$

$$\mu_{MKT}(t) - r = (1 - \gamma) \left(1 - \frac{MK \exp\{-r(T-t)\}}{X_{MKT}(t)}\right) \sigma_\delta^2. \quad (3.5)$$

Consequently: (i) $\mu_{MKT}(t) - r$ and $|\sigma_{MKT}(t)|$ are decreasing in M or K at a decreasing rate; (ii) $\mu_{MKT}(t) - r$ and $|\sigma_{MKT}(t)|$ are increasing in $\delta(t)$ or $X_{MKT}(t)$ for $M > 0$ and are independent of $\delta(t)$ or $X_{MKT}(t)$ for $M = 0$ (within an economy).

The market volatility and risk premium are stochastic and are driven by the level of the market and the amount invested in the bond to obtain the extra riskless wealth, $MK \exp\{-r(T-t)\}$.⁴ The market volatility and risk premium are both decreased by the presence of portfolio insurance. As seen in Lemma 2.3, the portfolio insurers invest a smaller fraction of their wealth in the risky asset and more in the bond, behaving as more risk-averse than the normal agents. So, to clear markets the risky asset must become more favorable as we introduce portfolio insurers into the economy. Since the interest rate and the market price of risk are unchanged, the only way to make the risky asset more favorable is for the market volatility to decrease. The market risk premium then decreases accordingly. This intuition is similar to that in Basak (1995), except that the extra demand for bond by the portfolio insurers now arises due to a riskless term in their wealth, rather than due to a put option.

Item (i) states that the effect of portfolio insurance on market volatility and risk premium is more pronounced for higher M . The more portfolio insurers there are, the higher the number of more risk averse agents to push the volatility down. Wealth effects (i.e., a dependence on each individual agent's endowment) are eliminated because of our choice of utility function. Consequently, more and sharper comparative statics may be derived than in Basak (1995) where wealth effects appear. Item (i) also states that the effect of portfolio insurance is more pronounced the higher the floor, since a higher floor leads to more risk-averse portfolio insurers. Result (ii) shows that the effect of portfolio insurance is less pronounced in "good" states where $\delta(t)$ and

⁴Since the portfolio insurers go long in the bond, in equilibrium the normal agents short the bond. However, the normal agents' wealth never becomes negative since the horizon market value does not go below the aggregate floor MK .

$X_{MKT}(t)$ are high, since here portfolio insurers do not have to strive as hard to meet their condition of horizon-wealth above the floor.

Remark 3.1 (Bond as the numeraire). If, alternatively, the bond price were used as a numeraire for the market price as in Grossman and Zhou (1996), all comparative statics results on the market price (Proposition 3.2) and its return dynamics (Proposition 3.3) remain valid.

4. Equilibrium in a Production Economy with Portfolio Insurance

Under pure-exchange, we have argued that the higher market price level under portfolio insurance arises to prevent agents from postponing consumption in aggregate, while the lower volatility prevents agents from net aggregate investment in the fixed (zero) supply bond. This motivates us to re-evaluate portfolio insurance under an economy in which both the aggregate consumption and the supply of bond are endogenous. We might expect the asset pricing results to break down under these circumstances. We choose a model to parallel the pure-exchange environment as closely as possible but with the bare minimum requirements of allowing consumption to actually be postponed in aggregate and allowing nonzero holdings in a riskless investment.

Accordingly, we adopt a variation on the Cox-Ingersoll-Ross (1985) continuous-time production economy. In this economy, the investment opportunities facing the agents are constant-returns-to-scale production technologies, using the single consumption good as their only input and producing the consumption good as output.⁵ If an amount q_i is invested in production technology i , the output at time dt later is $q_i + dq_i$ where the vector (2.1)

$$\begin{pmatrix} dq_0/q_0 \\ dq_1/q_1 \end{pmatrix} = \begin{pmatrix} r(t) dt \\ \mu(t) dt + \sigma(t) dW(t) \end{pmatrix}$$

is exogenous. The first production technology is riskless and the other risky, both with perfectly elastic supplies. In the production model, the parameters r, μ, σ of (2.1) are specified exogenously by the production technologies, and we take them to be constants. The agents' initial wealth, x_{n0} , is also exogenously specified as units of consumption good. For some recent applications using our type of production model with one riskless technology, see, for example, Constantinides (1990), Obstfeld (1994).⁶ In contrast, the Cox-Ingersoll-Ross (1985) model has one riskless bond

⁵According to our analysis in Section 2, we allow for a mechanism to "short-sell" these production technologies, either for an individual agent or in aggregate. To explicitly restrict such short-sales, we would have to apply constraints, $\phi_{ni}(t) \geq 0$, $i = 0, 1$, $n = 1, \dots, N$, to the agents' optimization problems. However, as long as we restrict the parameters, μ, σ, r , so that the optimal $\hat{\phi}_n$'s in (2.10)–(2.11) automatically obey these conditions, no agent's constraints are ever binding, and the solution is unaffected.

⁶For discrete-time, general equilibrium production models, see Prescott and Mehra (1980), Brock (1982), and Donaldson and Mehra (1984).

in zero net supply and no riskless production technology. Our intuition would suggest such a model might reveal similar results to ours, but the analysis becomes intractable because r is endogenously determined to be stochastic in the portfolio insurance economy.

The standard definition of equilibrium in production models of this nature is for all agents to be acting optimally and for all wealth to be invested in the two production technologies. However, to parallel our pure-exchange model, in Section 2 we have already applied the restriction to each individual agent that all his wealth be invested in the production technologies. Hence the only requirement for equilibrium is for each agent to be optimizing as defined in Section 2.

Definition 4.1. *An equilibrium in a production economy is a collection of optimal $(\hat{c}_n, \hat{\phi}_n, \hat{X}_n(T))$ for the specified (r, μ, σ) , satisfying*

$$\sum_{n=1}^N \hat{X}_n(t) \sum_{i=0}^1 \hat{\phi}_{ni}(t) = \sum_{n=1}^N \hat{X}_n(t) .$$

4.1. Equilibrium Aggregate Consumption

In the pure-exchange economy, as summarized by Proposition 3.1, the aggregate consumption supply is given exogenously by δ and the equilibrium ξ is determined by aggregating over all agents' first order conditions (2.5)–(2.6), and applying clearing in the consumption good. To determine equilibrium in the production economy, the agents' first order conditions are again used, but now ξ is the quantity given exogenously and the individual agents' consumption processes (and hence the aggregate consumption process) are determined endogenously. In this case it is convenient (and without loss of generality) to set $\xi(0) = 1$ across all economies. Then the agents' weights, y_m and y_n , are specified fully by equations (2.9), and we may not set $g(\Lambda)$ equal to 1. Proposition 4.1 presents conditions on and properties of the equilibrium aggregate consumption streams.

Proposition 4.1. *Equilibrium exists and equilibrium aggregate consumption, $\sum_{n=1}^N c_n$, in the production economy is given by*

$$\sum_{n=1}^N c_n(t) = \left[\frac{\xi(t)}{g(\Lambda)} \right]^{\frac{1}{\gamma-1}}, \quad (4.1)$$

where $\Lambda = (1/y_1, \dots, 1/y_N)$, $\bar{a}(t) \equiv [\exp\{\nu(T-t)\} - 1]/\nu \geq 0$, $\bar{b}(t) \equiv \exp\{\nu(T-t)\}$,

$$g(\Lambda)^{\frac{1}{1-\gamma}} = \frac{x_0 - M \exp\{-rT\}}{\bar{a}(0) + \bar{b}(0)}, \quad x_0 \equiv \sum_{n=1}^N x_{n0}, \quad \nu = \frac{\gamma}{1-\gamma} r + \frac{\gamma}{2(1-\gamma)^2} \theta^2 .$$

Consequently: (i) $\sum_{n=1}^N c_n(t)$ is decreasing in M or K .

We conclude that, in aggregate, in a portfolio insurance economy, pre-horizon consumption is actually foregone relative to a normal agent economy. Also the extent of this consumption postponement is increasing in the number of portfolio insurers present and in the level of the floor.

4.2. Market Price, Volatility and Risk Premium

The market, X_{MKT} , is defined as the aggregate wealth invested in the production technologies which, as in the pure-exchange economy, in equilibrium equals the sum of the N agents' wealth. Propositions 4.2 and 4.3 present the equilibrium market and dynamics in the production economy, in terms of exogenous quantities.

Proposition 4.2. *In equilibrium, the market price level is given by*

$$X_{MKT}(t) = g(\Lambda)^{\frac{1}{1-\gamma}} \bar{a}(t) \xi(t)^{\frac{1}{\gamma-1}} + g(\Lambda)^{\frac{1}{1-\gamma}} \bar{b}(t) \xi(t)^{\frac{1}{\gamma-1}} + MK \exp\{-r(T-t)\} ,$$

where $g(\Lambda)$, $\bar{a}(t)$ and $\bar{b}(t)$ are as given in Proposition 4.1. Consequently: (i) as $\xi(t) \rightarrow 0$, $X_{MKT}(t)$ is decreasing in M or K ; as $\xi(t) \rightarrow \infty$, $X_{MKT}(t)$ is increasing in M or K ; (ii) $X_{MKT}(t)$ is decreasing in $\xi(t)$ (within an economy).

In the production economy, we no longer have an unambiguous effect of portfolio insurance on the market value. In “bad” states ($\xi(t)$ high), the market value is higher in an economy with portfolio insurers than an economy with no portfolio insurers, but in “good” states ($\xi(t)$ low), the market value is lower. In either case, though, the effect of portfolio insurance increases in the number of portfolio insurers and in the level of the floor. The current value of the aggregate pre-horizon consumption, $g(\Lambda)^{1/(1-\gamma)} \bar{a}(t) \xi(t)^{1/(\gamma-1)}$, is now impacted (through $g(\Lambda)$) by portfolio insurance and is unambiguously decreasing in the number of portfolio insurers present and in the floor. This is because, while the valuation ξ of consumption is exogenously specified, aggregate pre-horizon consumption itself is foregone due to portfolio insurance. It is the impact of portfolio insurance on the current value of the horizon wealth (aggregate post-horizon consumption), $g(\Lambda)^{1/(1-\gamma)} \bar{b}(t) \xi(t)^{1/(\gamma-1)} + MK \exp\{-r(T-t)\}$, which is state-dependent; its dependence suggests that portfolio insurers shift post-horizon consumption from “bad” to “good” paths. As before, the portfolio insurance effect is independent of the individual agent's initial wealth; only the number of portfolio insurers matters. Property (ii) again shows that $X_{MKT}(t)$ is directly related to $1/\xi(t)$, as mentioned in Section 2.

Proposition 4.3 presents the market volatility and risk premium.

Proposition 4.3. *The equilibrium market volatility and risk are given by*

$$\begin{aligned} |\sigma_{MKT}(t)| &= \frac{1}{1-\gamma} \left(1 - \frac{MK \exp\{-r(T-t)\}}{X_{MKT}(t)} \right) \left| \frac{\mu - r}{\sigma} \right|, \\ \mu_{MKT}(t) - r &= \frac{1}{1-\gamma} \left(1 - \frac{MK \exp\{-r(T-t)\}}{X_{MKT}(t)} \right) \left(\frac{\mu - r}{\sigma} \right)^2, \end{aligned}$$

where $X_{MKT}(t)$ is as in Proposition 4.2. Consequently: (i) $|\sigma_{MKT}(t)|$ and $\mu_{MKT}(t) - r$ are lower for $M > 0$ than for $M = 0$; (ii) $|\sigma_{MKT}(t)|$ and $\mu_{MKT}(t) - r$ are decreasing in $\xi(t)$ for $M > 0$, and are independent of $\xi(t)$ for $M = 0$ (within an economy); (iii) $|\sigma_{MKT}(t)|$ and $\mu_{MKT}(t) - r$ are increasing in $X_{MKT}(t)$ for $M > 0$, and are independent of $X_{MKT}(t)$ for $M = 0$ (within an economy).

As in the pure-exchange economy, the market volatility and risk premium are reduced by the presence of portfolio insurance. The intuition in the production economy is again the higher effective risk aversion of the portfolio insurers relative to the normal agents. As seen in Lemma 2.3, the portfolio insurers have a lower demand for the risky opportunities than do the normal agents. Unlike in the pure-exchange economy, in this production economy the investment return parameters, r , μ , σ , are exogenous and so do not adjust to counteract the lower portfolio insurance demand. Instead, the actual aggregate investment in the risky production technologies is reduced compared with the investment in the riskless technology, and so the market becomes less risky, as reflected by the lower market volatility and risk premium.

Remark 4.1. We have also modeled portfolio insurance in a production model containing one firm, operating a non-constant-returns-to-scale technology, with labor as its input and consumption good as its output. If the consumer-workers derive additive log utility of consumption and leisure, we again deduce a reduced market volatility in the presence of portfolio insurance.

5. Trend-Chasing Behavior and Market Volatility

Portfolio insurance trading strategies are typically said to involve the transfer of funds to the risky assets as the value of the risky assets goes up, and the transfer of funds away from the risky assets as their value goes down. Critics argue that, as a result of this “trend-chasing” behavior, the market volatility will be increased by the presence of portfolio insurers. The connection between trend-chasing and instantaneous market volatility has also been discussed in the academic literature [Grossman (1988), Grossman and Zhou (1996)]. Hence in this section, we investigate whether our optimizing portfolio insurers of Section 2 are trend-chasers, and also any possible relationship between the presence of general trend-chasing portfolio insurers and

the market volatility. Accordingly, for both our optimizing portfolio insurance strategies and the prescribed strategies followed in practice, we study the sign of the (instantaneous) covariance of changes in risky security demand with changes in the risky security price, $d\hat{\alpha}_n(t) dP_1(t)$, where $\hat{\alpha}_n(t) \equiv \hat{\phi}_{n1}(t) X_n(t)/P_1(t)$ denotes the number of shares held in the stock. If $d\hat{\alpha}_n(t) dP_1(t) > 0$, we say that the strategy is “trend-chasing,” and if $d\hat{\alpha}_n(t) dP_1(t) < 0$, the strategy is “contrarian.” Such a (subjective) measure of trend-chasing or contrarian behavior in continuous-time has also been employed by Wang (1993, §6). Since in continuous-time, the portfolio holding $\alpha_n(t)$ is meant to represent the agent’s choice once he has observed the price $P_1(t)$, this measure does capture the notion of an agent responding to a price change.

For a tractable analysis of the prescribed trading strategies, in Subsections 5.1 and 5.2 we assume there are only two agents in the economy, and for brevity we use the pure-exchange environment. One agent is a portfolio insurer who follows some prescribed trading strategy and does not consume in $[0, T]$. (This assumption of no pre-horizon consumption of the portfolio insurer does not invalidate a comparison with our previous conclusions, because the effect of our previous portfolio insurers on the market price and volatility would have been the same if they had not consumed in $[0, T]$.) The other agent is a normal agent who has, for simplicity, log utility of consumption and log utility of horizon wealth.

5.1. Trend-Chasing Behavior of the Optimizing Portfolio Insurers

Trend-chasing behavior is a partial equilibrium notion, referring to the response of an agent’s demand to a price change, taking all other parameters as given, so for simplicity, we let the given price parameters, r , θ , μ and σ , be constants. From (2.10)–(2.11) and (A.2)–(A.3), it follows that for both logarithmic portfolio insurers and normal agents we have

$$d\hat{\alpha}_n(t) dP_1(t) = \frac{((T-t)+1)}{y_n \xi(t)} \theta[\theta - \sigma] dt, \quad n = 1, \dots, N.$$

The type of behavior of the agents depends on the market conditions (the sign of $\theta - \sigma$)⁷. We cannot argue that our portfolio insurers are unambiguously trend-chasers. For an identically endowed normal agent and portfolio insurer, $y_m > y_n$, so the portfolio insurer is less of a trend chaser than the normal agent.

5.2. Prescribed Trading Strategies Implementing Portfolio Insurance

Since our portfolio insurers cannot be argued to always act as trend-chasers, in order to address the viewpoint that a portfolio insurer should be defined as a trend-chaser, we now look at two

⁷For example, at the equilibrium price parameters with portfolio insurers present (Propositions 3.1 and 3.3), this sign is positive.

popular prescribed portfolio strategies: constant proportion portfolio insurance and portfolio insurance based on the synthetic put approach.

5.2.1. Constant Proportion Portfolio Insurance (CPPI)

The constant proportion portfolio insurance (CPPI) trading strategy invests in the risky asset a constant “multiple” of the difference between the agent’s wealth and some specified floor, and the remainder in the bond. (The floor is typically not a constant but grows at the risk-free rate.) Hence the proportion of wealth invested under the CPPI trading strategy is given by

$$\phi_{m1}(t) = \psi \left(1 - \frac{k \exp\{-r(T-t)\}}{X_m(t)} \right) ,$$

where $\psi > 1$ is the multiple, and $k \exp\{-r(T-t)\}$ is the floor. By comparison with (2.10), we note the close resemblance of our optimal portfolio insurance strategies to the CPPI trading strategy. The exogenous parameter ψ simply gets replaced by $\theta/(1-\gamma)\sigma$ (which is indeed larger than 1 at the equilibrium). For CPPI, we obtain

$$d\alpha_m(t) dP_1(t) = (\psi - 1)(X_m(t) - k \exp\{-r(T-t)\})\sigma^2 dt \geq 0 .$$

Hence under their perceived price processes (constant coefficients), the CPPI portfolio insurers are trend-chasers unless their wealth has hit the floor.

The assumptions of log utility and one risky asset allow us to readily derive an informative expression for the equilibrium market volatility. According to our earlier analysis (equations (2.6), (2.11), (A.3)), the normal agent’s optimal consumption, wealth and portfolio strategy are

$$\hat{c}_n(t) = \frac{1}{y_n \xi(t)} ; \quad \hat{X}_n(t) = \frac{T-t+1}{y_n \xi(t)} ; \quad \hat{\phi}_{n1} = \frac{\theta(t)}{\sigma(t)} ,$$

where $y_n = (T+1)/(\xi(0) x_{n0})$. Market clearing in the consumption good then implies

$$\xi(t) = \frac{1}{\delta(t)} , \quad r(t) = \mu_\delta - \sigma_\delta^2 \quad \text{and} \quad \theta(t) = \sigma_\delta .$$

Then a simple manipulation of the conditions from asset market clearing implies

$$|\sigma_{MKT}(t)| = \left| \frac{X_n(t)}{X_n(t) + [X_m(t) - \psi(X_m(t) - k \exp\{-r(T-t)\})]} \right| |\sigma_\delta| .$$

In a benchmark economy with all normal agents we have $|\sigma_{MKT}(t)| = |\sigma_\delta|$. Whether the volatility is lower or higher in the portfolio insurance economy depends on the term $X_m(t) - \psi(X_m(t) - k \exp\{-r(T-t)\})$, which corresponds to the amount of wealth the portfolio insurer invests in the bond. If the portfolio insurer holds a positive amount in the bond, the market volatility is decreased compared with the benchmark economy; if he has a small negative amount in the bond, the volatility is increased; and if he has a large enough (relative to $X_n(t)$) negative amount in the bond, the volatility is again decreased. In conclusion, the CPPI agents are unambiguously trend-chasers, but the market volatility may increase or decrease in their presence.

5.2.2. Synthetic Put Approach to Portfolio Insurance

The idea behind the synthetic put approach is that portfolio insurers desire a payoff at T from their portfolio of the form $\max\{K, \pi P_1(T)\}$, where $\pi > 0$. Under the Black-Scholes (1973) environment, when the stock price follows a geometric Brownian motion (and the stock pays no dividends), the following strategy is self-financing and attains the required payoff:

$$\phi_{m1}(t) = \frac{\pi \mathcal{N}(d_1) P_1(t)}{X_m(t)} > 0, \quad \phi_{m0}(t) = \frac{K \exp\{-r(T-t)\} \mathcal{N}(-d_2)}{X_m(t)} > 0,$$

where $d_1 \equiv \frac{\ln(P_1(t)/K) + (r + \sigma^2/2)(T-t)}{|\sigma| \sqrt{T-t}}$, $d_2 = d_1 - |\sigma| \sqrt{T-t}$. We derive

$$d\alpha_m(t) dP_1(t) = \frac{\pi \mathcal{N}'(d_1)}{\sqrt{T-t}} |\sigma| P_1(t) dt > 0.$$

Hence under their perceived Black-Scholes environment, the portfolio insurers are trend-chasers. By imposing market clearing and manipulating as in Section 5.2.1, we can express the market volatility in the portfolio insurance economy as

$$|\sigma_{MKT}(t)| = \left(\frac{\hat{X}_n(t)}{\hat{X}_n(t) + K \exp\{-r(T-t)\} \mathcal{N}(-d_2)} \right) |\sigma_\delta|,$$

always lower than the volatility in the benchmark economy. Hence, although these agents are trend-chasers, the market volatility is always decreased by their presence.

6. Conclusion

We have carried out a comparative analysis of the effects of portfolio insurance on the general equilibrium market price and dynamics. Portfolio insurers are modeled as expected utility maximizers who derive utility from consumption in $[0, T]$ and from wealth at some horizon T . The utility of horizon wealth is chosen to have its marginal utility going smoothly to infinity at the floor. Thus the portfolio insurers become gradually, rather than suddenly, unhappy as the wealth tends towards the floor; this contrasts with Grossman and Vila (1989), Basak (1995), Grossman and Zhou (1996).⁸ This formulation provides considerable tractability over a formulation with an explicit constraint, and is seen to resemble CPPI. We investigate and compare the effects of portfolio insurance in a pure-exchange and a production model. Some of the results of Basak (1995) are reinforced in both models; that the market volatility and risk premium are decreased by the presence of portfolio insurance.

⁸For an analysis of a “softer” portfolio insurance constraint, permitting the horizon wealth to deteriorate below the floor with a prespecified probability, see the Value-at-Risk constraint analyzed by Basak and Shapiro (2001).

The analysis reveals that it is primarily in the price and consumption valuation levels that the pure-exchange and production models differ from each other. In pure-exchange, the exogenous aggregate consumption may not be impacted by portfolio insurance, but the Arrow-Debreu prices adjust to increase the value of some post-horizon consumption relative to pre-horizon consumption. In this case, the reason for the reduced market volatility is that prices adjust to counteract the portfolio insurer's tendency to hold less risky portfolios. In the production case, the Arrow-Debreu prices are specified exogenously, and the now-endogenous aggregate consumption itself is affected by portfolio insurance. The portfolio insurers postpone some consumption until after the horizon, and also actually shift horizon wealth from "good" paths where it would well-exceed the floor, to "bad" paths where the horizon wealth would fall closer to the floor. Hence the reduced market volatility, revealing the mechanism to be different from the pure-exchange economy. Using the portfolio insurance application, the analysis demonstrates more generally the critical differences between a pure-exchange and a production model.

Appendix: Proofs

Proof of Lemma 2.1: See, for example, Lemma 2.4, Cox and Huang (1989). *Q.E.D.*

Proof of Lemma 2.2: (2.5)–(2.9) are derived by standard static optimization techniques [e.g., Cox and Huang (1989)]. Property (i) is immediate from (2.9). *Q.E.D.*

Proof of Lemma 2.3: For any optimally invested wealth $\hat{X}(t)$ satisfying (2.3) we have

$$d(\xi(t) \hat{X}(t)) + \xi(t) \hat{c}(t) dt = \xi(t) \hat{X}(t) [\hat{\phi}_1(t) \sigma(t) - \theta(t)] dW(t), \quad (\text{A.1})$$

by applying Itô's Lemma to the product $\xi(t) \hat{X}(t)$. From Lemma 2.1 and the optimal consumptions and horizon wealth in Lemma 2.2, the optimally invested wealth of the agents is

$$\begin{aligned} \hat{X}_m(t) &= \frac{y_m^{\frac{1}{\gamma-1}}}{\xi(t)} E \left[\int_t^T \xi(s)^{\frac{\gamma}{\gamma-1}} ds \mid \mathcal{F}_t \right] + \frac{y_m^{\frac{1}{\gamma-1}}}{\xi(t)} E \left[\xi(T)^{\frac{\gamma}{\gamma-1}} \mid \mathcal{F}_t \right] + \frac{K}{\xi(t)} E[\xi(T) \mid \mathcal{F}_t], \\ \hat{X}_n(t) &= \frac{y_n^{\frac{1}{\gamma-1}}}{\xi(t)} E \left[\int_t^T \xi(s)^{\frac{\gamma}{\gamma-1}} ds \mid \mathcal{F}_t \right] + \frac{y_n^{\frac{1}{\gamma-1}}}{\xi(t)} E \left[\xi(T)^{\frac{\gamma}{\gamma-1}} \mid \mathcal{F}_t \right]. \end{aligned}$$

$\xi(t)$ follows a geometric Brownian motion, so $\{\mathcal{F}_t\} = \{\xi(t)\}$ and, given $\xi(t)$, $\ln \xi(s)$, $s \in [t, T]$ is normally distributed with mean $\ln \xi(t) - r(s-t) - \|\theta\|^2(s-t)/2$ and variance $\theta^2(s-t)$. $\ln \xi(s)^{\gamma/(\gamma-1)}$ is also normally distributed, allowing the expectations to be evaluated as

$$\hat{X}_m(t) = y_m^{\frac{1}{\gamma-1}} \bar{a}(t) \xi(t)^{\frac{1}{\gamma-1}} + y_m^{\frac{1}{\gamma-1}} \bar{b}(t) \xi(t)^{\frac{1}{\gamma-1}} + K \exp\{-r(T-t)\}, \quad m = 1, \dots, M, \quad (\text{A.2})$$

$$\hat{X}_n(t) = y_n^{\frac{1}{\gamma-1}} \bar{a}(t) \xi(t)^{\frac{1}{\gamma-1}} + y_n^{\frac{1}{\gamma-1}} \bar{b}(t) \xi(t)^{\frac{1}{\gamma-1}}, \quad n = M+1, \dots, N, \quad (\text{A.3})$$

where $\bar{a}(t) \equiv \frac{1}{\rho} [\exp\{\rho(T-t)\} - 1] \geq 0$, $\bar{b}(t) \equiv \exp\{\rho(T-t)\}$, $\rho \equiv \frac{\gamma}{1-\gamma} r + \frac{\gamma}{2(1-\gamma)^2} \theta^2$.

Applying Itô's Lemma to $\xi(t) \hat{X}_m(t)$ and $\xi(t) \hat{X}_n(t)$, then yields

$$\begin{aligned} d(\xi(t) \hat{X}_m(t)) &= y_m^{\frac{1}{\gamma-1}} \bar{a}(t) \frac{\gamma}{1-\gamma} \xi(t)^{\frac{\gamma}{\gamma-1}} \theta(t) dW(t) + y_m^{\frac{1}{\gamma-1}} \bar{b}(t) \frac{1}{1-\gamma} \xi(t)^{\frac{\gamma}{\gamma-1}} \theta(t) dW(t) \\ &\quad - K \exp\{-r(T-t)\} \xi(t) \theta(t) dW(t) + dt \text{ terms} \\ &= \xi(t) \left[\frac{\gamma}{1-\gamma} \hat{X}_m(t) - \frac{1}{1-\gamma} K \exp\{-r(T-t)\} \right] \theta(t) dW(t) + dt \text{ terms}, \\ d(\xi(t) \hat{X}_n(t)) &= y_n^{\frac{1}{\gamma-1}} \bar{a}(t) \frac{1}{1-\gamma} \xi(t)^{\frac{\gamma}{\gamma-1}} \theta(t) dW(t) + y_n^{\frac{1}{\gamma-1}} \bar{b}(t) \frac{1}{1-\gamma} \xi(t)^{\frac{\gamma}{\gamma-1}} \theta(t) dW(t) + dt \text{ terms} \\ &= \xi(t) \hat{X}_n(t) \theta(t) dW(t) + dt \text{ terms}. \end{aligned}$$

Equating the $dW(t)$ terms with those of equation (A.1) yields (2.10) and (2.11). Furthermore, since the numerators and denominators of both fractions are positive, we have

$$1 > \left[1 - \frac{K \exp\{-r(T-t)\}}{\hat{X}_m(t)} \right] = \left[\frac{y_m^{\frac{1}{\gamma-1}} \bar{a}(t) \xi(t)^{\frac{1}{1-\gamma}} + y_m^{\frac{1}{\gamma-1}} \bar{b}(t) \xi(t)^{\frac{1}{1-\gamma}}}{\hat{X}_m(t)} \right] > 0.$$

Hence $\hat{\phi}_{m1}(t) \leq \hat{\phi}_{n1}(t)$. *Q.E.D.*

Proof of Proposition 3.1: Existence of equilibrium is established by showing existence of r, μ, σ , whose associated ξ and optimal policies clear all markets. $\hat{c}_m$ and $\hat{c}_n$ given by substituting (3.2) into (2.5), (2.6), (2.9) exist since the expectations (and hence agents' weights y_n, y_m) are well-defined for δ a geometric Brownian motion, and also imply the first equality in (3.1). The proof that the first equality in (3.1) implies the remaining appears in Basak (1995). Existence of r, μ, σ whose corresponding ξ is as given, is demonstrated by the explicit formulae in Propositions 3.1 and 3.3, which also reveal that r, μ and σ are bounded, with σ bounded away from zero. To show that the equilibrium state price density is necessarily of form (3.2), observe that $\hat{c}_m$ and $\hat{c}_n$ given by (2.5), (2.6), (2.9), together with (3.1) imply (3.2), making use of the standard properties of a representative agent's utility function. Finally, applying Itô's Lemma to (3.2) yields r and θ . *Q.E.D.*

Proof of Proposition 3.2: $X_{MKT}(t)$ in (3.3) is determined by aggregating over agents' optimal wealth given by equations (A.2)–(A.3) in the proof of Lemma 2.3, recalling $g(\Lambda) = 1$ and substituting for the equilibrium state price density $\xi(t)$ from (3.2). Properties (i) and (ii) are immediate. *Q.E.D.*

Proof of Proposition 3.3: Applying Itô's Lemma to the market expression in Proposition 3.2 yields (3.4)–(3.5). Substituting for $X_{MKT}(t)$ and taking derivatives yields (i) and (ii). *Q.E.D.*

Proof of Proposition 4.1: Summing (2.5)–(2.6) over agents and using the representative agent with $\lambda_n = 1/y_n$, $n = 1, \dots, N$, we obtain (4.1). The restriction on $g(\Lambda)$ is obtained by summing (2.9). Result (i) is straightforward to derive, after substituting $g(\Lambda)$ into (4.1). Existence is established as in Proposition 3.1. *Q.E.D.*

Proof of Proposition 4.2: Summing (A.2)–(A.3) over agents yields $X_{MKT}(t)$. As $\xi(t) \rightarrow 0$,

$$X_{MKT}(t) \rightarrow g(\Lambda)^{\frac{1}{1-\gamma}} \bar{a}(t) \xi(t)^{\frac{1}{\gamma-1}} + g(\Lambda)^{\frac{1}{1-\gamma}} \bar{b}(t) \xi(t)^{\frac{1}{\gamma-1}},$$

yielding the first part of (i), since $g(\Lambda)^{1/1-\gamma}$ is decreasing in M and K . As $\xi(t) \rightarrow \infty$, we obtain

$$X_{MKT}(t) \rightarrow MK \exp\{-r(T-t)\},$$

yielding the second part of (i). Result (ii) is immediate from the expression for $X_{MKT}(t)$. *Q.E.D.*

Proof of Proposition 4.3: Applying Itô's Lemma to $X_{MKT}(t)$ in Proposition 4.2 yields $|\sigma_{MKT}(t)|$ and $\mu_{MKT}(t) - r$. Then (i) and (iii) are immediate. Substituting back in for $X_{MKT}(t)$, we obtain

$$|\sigma_{MKT}(t)| = \frac{1}{1-\gamma} \left(\frac{g(\Lambda)^{\frac{1}{1-\gamma}} \bar{a}(t) \xi(t)^{\frac{1}{\gamma-1}} + g(\Lambda)^{\frac{1}{1-\gamma}} \bar{b}(t) \xi(t)^{\frac{1}{\gamma-1}}}{g(\Lambda)^{\frac{1}{1-\gamma}} \bar{a}(t) \xi(t)^{\frac{1}{\gamma-1}} + g(\Lambda)^{\frac{1}{1-\gamma}} \bar{b}(t) \xi(t)^{\frac{1}{\gamma-1}} + MK \exp\{-r(T-t)\}} \right) \left| \frac{\mu - r}{\sigma} \right|,$$

and then $\mu_{MKT}(t) - r = |\sigma_{MKT}(t)|$, from which we derive (ii). *Q.E.D.*

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